

FAIR-PLACO: Towards Fairness-Aware Human-AI Team Formation

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1 Introduction

Human-AI collaborative systems have become essential in high-stakes domains such as healthcare, criminal justice, and credit assessment, where humans and AI classifiers each cover different failure modes. COMHAI formalised this for the multi-human setting via a Bayesian combination rule, and proved that including all humans can *decrease* accuracy — making intelligent subset selection necessary. PLACO pushed further by estimating human labels before querying, enabling cost-effective selection under a budget constraint.

Despite these advances, both frameworks are fairness-blind. A subset-selection policy that maximises expected accuracy can still systematically under-serve minority groups or amplify biases in the AI model. Learning-to-defer systems can magnify group disparities when fairness is absent from the routing objective. FAIR-PLACO fills this gap, guided by three design goals:

1. Combine humans and AI to improve both accuracy and fairness.
2. Handle multiple humans with heterogeneous costs.
3. Remain robust when humans themselves exhibit bias.

The remainder is structured as follows: §2 reviews the base frameworks; §3 defines the metrics; §4 describes FAIR-PLACO; §5 covers the experimental setup; §6 presents results; §8 discusses findings and limitations; §9 provides a radar style, per-dataset performance analysis; §10 concludes.

2 Background

2.1 ComHAI: Combining Multiple Humans with AI

Consider a K -way classification problem where an AI model $\mathcal{M}$ produces a calibrated probability vector $\mathbf{m}^\theta(x) \in \Delta^K$, and n humans each provide a hard label $h_i(x) \in \{1, \dots, K\}$. Each human’s reliability is encoded by a $K \times K$ confusion matrix $\phi^{[i]}$, where $\phi_{st}^{[i]} = P(h_i(x) = s \mid y(x) = t)$, estimated with a Dirichlet prior.

Applying Bayes’ rule under the assumption that human predictions are conditionally independent given the true label:

$$P(y(x) = j \mid \mathbf{h}(x), \mathbf{m}(x)) \propto m_j^\theta(x) \prod_{i=1}^n \phi_{h_i(x), j}^{[i]}. \quad (1)$$

A key theoretical result (the “Too Many Cooks” lemma) shows that this posterior is *non-monotone* in the number of humans included: adding a human whose information ratio $\phi_{h_i(x), y(x)}^{[i]} / (1 - \phi_{h_i(x), y(x)}^{[i]}) < 1$ can hurt accuracy. The *GreedySubsetSelection* algorithm therefore includes only humans for whom that ratio exceeds 1, using a pseudo-label in place of the unknown true label.

2.2 PLACO: Cost-Effective Human Subset Selection

It is that COMHAI requires each human’s true label on every instance to compute the information ratio, an assumption that is expensive in practice. PLACO decouples *label estimation* from *label elicitation*:

1. **Step 1 — Label estimation.** Before querying any human, estimate what each human i would say:

$$\hat{h}_i(x) = \arg \max_{l \in \mathcal{Y}} \sum_{y \in \mathcal{Y}} \phi_{ly}^{[i]} m_y(x). \quad (2)$$

2. **Step 2 — Value function and subset selection.** Using these estimates, compute a piecewise value $V_i(x, j)$ derived from a lower bound on the accuracy ratio; select humans whose value exceeds 1; elicit true labels only from those selected.

Two solvers are provided: a greedy variant (*PLACO Greedy*) and an Integer Linear Program (*PLACO LP*) that enforces a hard budget constraint B .

3 Fairness Definitions

Let $\hat{Y}$ denote the combined model’s prediction, Y the true label, and $A \in \{0, 1\}$ the sensitive attribute ($A = 1$ = privileged group). Seven metrics are tracked throughout this work.

Definition 1 (Demographic Parity (DP)). $DP = |P(\hat{Y} = 1 \mid A = 1) - P(\hat{Y} = 1 \mid A = 0)|$. Measures whether both groups receive positive predictions at the same rate. Lower is fairer; ideal value is 0.

Definition 2 (Equal Opportunity (EOpp)). $EOpp = |\text{TPR}_1 - \text{TPR}_0|$ where $\text{TPR}_a = P(\hat{Y} = 1 \mid Y = 1, A = a)$. Ensures qualified individuals in both groups have equal recall. Lower is fairer.

Definition 3 (Equalized Odds (EO)). $EO = \max(|\text{TPR}_1 - \text{TPR}_0|, |\text{FPR}_1 - \text{FPR}_0|)$. Both true-positive and false-positive rates must match across groups. Lower is fairer.

Definition 4 (Treatment Equality (TE)). $TE = \left| \log \frac{\text{FN}_0 + 1}{\text{FP}_0 + 1} - \log \frac{\text{FN}_1 + 1}{\text{FP}_1 + 1} \right|$ (Berk et al., 2018). A high TE gap means the *type* of error differs systematically by group. Lower is fairer.

Definition 5 (Subgroup Fairness). Worst-case TPR gap across intersections of the sensitive attribute and feature quantile buckets. Captures finer-grained disparities not visible at the top-level group. Lower is fairer.

Definition 6 (Individual Fairness (Consistency)). $IF = \frac{1}{n} \sum_{i=1}^n \text{avg}_{j \in \text{kNN}(i)} \mathbf{1}[\hat{y}_j \neq \hat{y}_i]$. Similar inputs should receive similar decisions. Higher is better; range $[0, 1]$.

4 The FAIR-PLACO Framework

FAIR-PLACO extends PLACO along two complementary dimensions: (1) a fairness-aware AI model used as the probabilistic prior, and (2) a Lagrangian-penalised value

function that discourages human routing decisions that worsen running group disparities.

4.1 Fairness-Aware AI (FairAI)

FAIRAI replaces the unconstrained baseline classifier with `ExponentiatedGradient(EqualizedOdds, HistGradientBoostingClassifier)` from Fairlearn. This reduction enforces a Hardt-style equalized-odds bound at training time, producing a prior $\mathbf{m}(x)$ with a smaller pre-routing fairness gap. This lever is *independent* of the Lagrangian routing — both can be ablated and combined separately.

4.2 Human Simulator Regimes

Three behaviour models are studied with a pool of $n = 15$ humans (accuracies $\alpha_j \in [0.29, 0.94]$, reflecting realistic annotator diversity):

Unbiased. Human j flips the true label uniformly with probability $1 - \alpha_j$. No dependence on the instance’s sensitive group.

Explicitly Biased. When the AI is uncertain (confidence $\in [0.40, 0.60]$) and the instance is unprivileged ($A = 0$): even-indexed humans issue a hard-denial ($\hat{h} = 0$), while odd-indexed ones apply a soft accuracy penalty $\tilde{\alpha}_j = \max(0, \alpha_j - 0.30)$ on minority instances.

Automation-Biased. Each human copies the AI’s hard label with probability $\tau = 0.85$, falling back to their own accuracy with probability $1 - \tau$.

4.3 The Lagrangian Value Function

Let Ψ_m denote the running violation of fairness metric $m \in \{\text{dp}, \text{eopp}, \text{eo}, \text{te}\}$ after processing some prefix of the test set, and let $\Delta\Psi_m(i, S)$ denote the *predicted marginal increment* in Ψ_m if human subset S handles instance i . The original PLACO value $V^{\text{PLACO}}(S)$ is a product of per-human information ratios and grows exponentially with $|S|$. To prevent the accuracy term from dominating the penalty at any subset size, we work in *log space*:

$$\begin{aligned} \log V^{\text{fair}}(S, i) = & -K \underbrace{\sum_m \lambda_m \max(0, \Delta\Psi_m(i, S))}_{\text{marginal penalty}} \\ & - \underbrace{0.1K \sum_m \lambda_m \Psi_m}_{\text{state penalty}} - \underbrace{\beta C(S, i)}_{\text{cost}} + \underbrace{\log V^{\text{PLACO}}(S)}_{\text{accuracy term}} \end{aligned} \quad (3)$$

where $K = 200$ is a *penalty strength* that calibrates the $O((1/n)^2)$ violation increments to the same log-scale as $\log V^{\text{PLACO}}$; $\lambda_m \geq 0$ are dual weights; β is a cost coefficient; and $C(S, i)$ is the total cost of the chosen subset.

The marginal penalty $\Delta\Psi_m(i, S)$ is computed from a `FairnessState` object that maintains soft running sums (predicted positive probabilities, soft true-positive counts, soft false-positive counts, etc.) without requiring the unknown true label. Specifically:

$$\text{DP violation} = \max(0, |q_0 - q_1| - \varepsilon)^2 \quad (4)$$

$$\text{EOpp violation} = \max(0, |\text{TPR}_0 - \text{TPR}_1| - \varepsilon)^2 \quad (5)$$

$$\text{EO violation} = (\text{TPR}_0 - \text{TPR}_1)^2 + (\text{FPR}_0 - \text{FPR}_1)^2 \quad (6)$$

where $p_{y_t} = m_1(x_t)$ is the AI’s soft positive probability and $\hat{q}_t$ is the combined system’s predicted positive probability for instance t ; $\varepsilon = 0.05$ is an allowed slack.

4.4 Online Dual Ascent

The dual weights $\{\lambda_m\}$ are updated every $B_{\text{batch}} = 64$ instances using a projected gradient step:

$$\lambda_m \leftarrow \text{clip}(\lambda_m + \eta(\Gamma_m - 0), 0, \lambda_{\text{max}}), \quad (7)$$

where Γ_m is the realised violation on the most recent batch, $\eta = 0.5$ is the step size, and $\lambda_{\text{max}} = 50$. As a violation grows, its penalty weight grows; if it falls below ε the weight decays. This drives the system toward near-feasibility without fixing a rigid budget on fairness.

4.5 The FAIR-PLACO Policy

At each test instance x_i , FAIR-PLACO solves a small Integer Linear Program using the singleton Lagrangian values as coefficients:

$$\begin{aligned} \max_{e_h \in \{0,1\}} \quad & \sum_{h=1}^n V^{\text{fair}}(\{h\}, i) \cdot e_h \\ \text{subject to} \quad & \sum_{h=1}^n c_{i,h} e_h \leq B_{\text{instance}}, \end{aligned} \quad (8)$$

where $B_{\text{instance}} = B_{\text{global}}/n_{\text{test}}$ distributes the total budget equally across instances. The true labels of the chosen subset are then elicited and combined with the AI probabilities via Equation (1). After each instance the fairness state is updated; after every B_{batch} instances the dual weights are ascended.

5 Experimental Setup

5.1 Datasets

Table 1: Dataset summary. Sensitive attributes and positive-outcome definitions.

Dataset	Sensitive attr.	Privileged ($A = 1$)	Positive outcome	Size
COMPAS	Race	Caucasian	Did <i>not</i> reoffend	6 172
Adult	Sex	Male	Income >50K	45 222
German	Sex	Male	Good credit	1 000
ACS Income	Race	White	Income >\$50K	15 000
CIFAR-10H	AI confidence	High-confidence	Label ≥ 5 (binarised)	10 000
ImageNet-16H	AI confidence	High-confidence	Label ≥ 8 (binarised)	varies

All datasets use a 70/30 stratified split with standardised features. For the image datasets, labels are binarised at the median class index and the sensitive attribute is synthesised from the AI’s top predicted probability (high vs. low confidence), serving as a proxy for easy vs. hard cases.

5.2 The 12-Setting Experiment Grid

Table 2: The twelve experimental settings. Group 3 (settings g–i) is the FAIR-PLACO main contribution; Group 4 (j–l) is the ablation.

Code	AI Model	Policy	Human Bias
<i>Group 1 — Baseline (accuracy-only)</i>			
a	BasicAI	PLACO	Unbiased
b	BasicAI	PLACO	Explicitly biased
c	BasicAI	PLACO	Automation biased
<i>Group 2 — Fair AI prior only</i>			
d	FairAI	PLACO	Unbiased
e	FairAI	PLACO	Explicitly biased
f	FairAI	PLACO	Automation biased
<i>Group 3 — FAIR-PLACO (main contribution)</i>			
g	FairAI	FAIR-PLACO	Unbiased
h	FairAI	FAIR-PLACO	Explicitly biased
i	FairAI	FAIR-PLACO	Automation biased
<i>Group 4 — Ablation (Lagrangian only, no fair AI)</i>			
j	BasicAI	FAIR-PLACO	Unbiased
k	BasicAI	FAIR-PLACO	Explicitly biased
l	BasicAI	FAIR-PLACO	Automation biased

Hyperparameters. Budget fraction = 0.30; dual step $\eta = 0.5$; fairness slack $\varepsilon = 0.05$; penalty strength $K = 200$; $\lambda_{\text{init}} = 1.0$; $\lambda_{\text{max}} = 50$; batch size = 64; 15 humans per run. Per-instance per-human costs are sampled from $\mathcal{U}(0.0001, 2)$.

- **Group 1 (a–c) — The unmodified baseline.** This is standard human-AI teaming with no fairness awareness anywhere in the pipeline. It establishes what you naturally get when you optimise purely for accuracy, serving as the reference point every other group is measured against.
- **Group 2 (d–f) — Fix the AI, leave the routing alone.** The only change is training the AI under an equalized-odds constraint. This isolates how much of the fairness problem can be solved at the model level, before any human is even consulted — a pre-emptive, upstream intervention.
- **Group 3 (g–i) — The full proposed system.** Both levers are on: a fair AI prior and fairness-aware human routing. This is the complete FAIR-PLACO framework and represents the claim that fairness should be addressed at every stage — what the AI believes going in, and who gets asked to weigh in.
- **Group 4 (j–l) — Can smart routing compensate for an unfair AI?** The Lagrangian routing is active, but the underlying AI is unconstrained. This tests whether the human selection policy alone can recover fairness that the model never had — isolating the routing’s standalone contribution and revealing its limits when it has no fair prior to build on.

6 Results

Table 3 displays (model $\times$ metric) performance outputs, averaged over all datasets.

- Accuracy degrades clearly as human bias increases — automation bias is the harshest regime across all four model groups, dropping team accuracy by over 15 percentage points.
- Groups 2 and 3 (FAIRAI-based) consistently show smaller fairness gaps in the unbiased column despite near-identical accuracy to Group 1 — the fair AI prior is carrying most of this improvement.
- FAIR-PLACO’s Lagrangian routing (Group 3 vs. Group 2) provides an additional reduction in EO and DP under unbiased humans, visible as a further narrowing of the right-axis fairness bars.

7 Win Count Analysis

Table 4 summarises how many (dataset $\times$ metric) cells each setting achieves rank 1 on.

- Setting d (PLACO + FAIRAI, unbiased) leads with 16 wins, driven primarily by dominating EO across the fairness datasets — the fair AI prior alone is the single most effective lever.
- Setting g (FAIR-PLACO + FAIRAI, unbiased) follows with 12 wins, outperforming d on EOpp in two additional cells — the Lagrangian contributes targeted, metric-specific gains on top of the prior.
- No single setting wins on accuracy and all fairness metrics simultaneously; the Lagrangian routing is complementary to rather than a replacement for a fair AI prior.

8 Discussion

8.1 Limits of Fairness Routing Under Biased Humans

A consistent negative finding across all settings is that human bias dominates fairness outcomes when present. Under explicit bias, EO gaps grow from ≈ 0.02 (unbiased) to ≈ 0.20 — an order-of-magnitude increase that neither FairAI nor FAIR-PLACO can fully counteract. This occurs because the Lagrangian penalises the *marginal increment* in fairness violations but cannot avoid selecting at least one human per instance (the ILP always selects the maximum-valued human), and all available humans are systematically biased.

8.2 Automation Bias: A Nuanced Case

Automation-biased humans copy the AI’s hard label 85% of the time. When the AI itself is fair (settings f and i), this effectively amplifies the AI’s fairness properties. This explains why settings f and i show the lowest DP gaps in some datasets (e.g., CIFAR-10H DP ≈ 0.041). Conversely, when the AI is not fair (settings c, l), automation bias propagates its unfairness. This suggests a novel design insight: automation bias can be a feature rather than a bug when paired with a sufficiently fair AI prior.

8.3 Treatment Equality: Calibration Challenge

TE results are difficult to interpret because the Lagrangian penalises a normalised soft rate (FNR/n, FPR/n) that stays in the $O((1/n)^2)$ range, while the

Table 3: Average metrics across all six datasets for the 12 settings. Best per column **bolded**. Acc. $\uparrow$; all gap metrics $\downarrow$; IndivF. $\uparrow$.

Setting	Acc.	DP $\downarrow$	EO $\downarrow$	EOpp $\downarrow$	TE $\downarrow$	Subgrp. $\downarrow$	IndivF. $\uparrow$
a) PLACO + BasicAI + Unbiased	0.970	0.096	0.023	0.015	0.243	0.067	0.700
b) PLACO + BasicAI + Explicit	0.884	0.091	0.197	0.162	0.309	0.322	0.719
c) PLACO + BasicAI + Auto	0.769	0.107	0.156	0.122	0.382	0.436	0.847
d) PLACO + FairAI + Unbiased	0.969	0.095	0.019	0.012	0.228	0.069	0.698
e) PLACO + FairAI + Explicit	0.878	0.082	0.199	0.157	0.537	0.319	0.708
f) PLACO + FairAI + Auto	0.773	0.074	0.122	0.088	0.230	0.394	0.840
g) FAIR-PLACO + FairAI + Unbiased	0.967	0.093	0.022	0.013	0.299	0.079	0.698
h) FAIR-PLACO + FairAI + Explicit	0.878	0.081	0.199	0.158	0.554	0.323	0.706
i) FAIR-PLACO + FairAI + Auto	0.772	0.074	0.122	0.087	0.228	0.393	0.841
j) FAIR-PLACO + BasicAI + Unbiased	0.968	0.094	0.024	0.017	0.293	0.078	0.700
k) FAIR-PLACO + BasicAI + Explicit	0.883	0.089	0.198	0.163	0.332	0.324	0.719
l) FAIR-PLACO + BasicAI + Auto	0.769	0.107	0.155	0.121	0.379	0.435	0.847

Table 4: Number of rank-1 finishes across 6 datasets $\times$ 7 metrics = 42 cells (top 5 settings only; remaining 7 scored zero wins).

Setting	Acc	DP	EO	EOpp	TE	Subgrp	IF	Total
a) PLACO + Basic + Unbiased	6	0	0	1	0	4	0	11
d) PLACO + Fair + Unbiased	3	1	5	3	1	3	0	16
g) FAIR-PLACO + Fair + Unbiased	1	1	4	4	0	2	0	12
j) FAIR-PLACO + Basic + Unbiased	3	0	0	0	2	3	0	8
i) FAIR-PLACO + Fair + Auto	0	3	0	0	1	0	2	6

post-hoc evaluator uses a log-ratio of raw counts. A single FN imbalance in a small group can cause the evaluator to report a large TE gap even when the Lagrangian considers it small. Future work should align the penalty and evaluator formulations more closely.

8.4 Shortcomings and Directions for Future Work

1. **No feedback loop.** Human confusion matrices are estimated once at training time and held fixed. In practice, human behaviour adapts (e.g., confirmation bias may grow over time), which would require online updating of confusion matrices.
2. **Random cost function.** Costs are sampled from $\mathcal{U}(0.0001, 2)$, a worst-case scenario with no structure. Real-world costs correlate with human expertise (higher-accuracy annotators cost more). A cost model that reflects this could enable Pareto-optimal accuracy-cost-fairness trade-offs.
3. **Binary classification only.** Extending to K -way classification (as in the original ComHAI) with multi-category sensitive attributes (e.g., race with multiple categories) is a natural next step.
4. **Lagrangian instability under small datasets.** German Credit (1000 instances) is the smallest dataset tested. The dual ascent updates every 64 instances, giving only ≈ 10 gradient steps over the test set, which is insufficient for convergence. Adaptive batch sizing could help.
5. **Intersectional fairness.** The current framework treats A as binary. Real applications may require simultaneous enforcement of fairness across race, gen-

der, and age, which would require multi-dimensional constraint tracking.

9 Per-Dataset Radar Charts (Unbiased Human Setting)

Figure 1 displays per-dataset radar charts for model settings with unbiased humans (a, d, g, j) over various performance metrics.

- Setting j (Lagrangian routing, no fair AI prior) collapses sharply on the TE and Subgroup axes on Adult and ACS Income while a, d, and g maintain their shape — routing alone not only fails to improve subgroup fairness but actively worsens it relative to the plain baseline.
- Settings d and g are visually indistinguishable across all six datasets — their polygons overlap almost perfectly, confirming the fair AI prior does the heavy lifting and the Lagrangian adds negligible shape change on top of it.
- German Credit shows the widest four-way spread and is the only dataset where all four settings take meaningfully different shapes, making it the clearest illustration of the accuracy-fairness tension.
- COMPAS, CIFAR-10H, and ImageNet-16H show near-identical polygons for all settings — the fairness interventions have no visible traction where the sensitive attribute does not create sufficient group disparity.

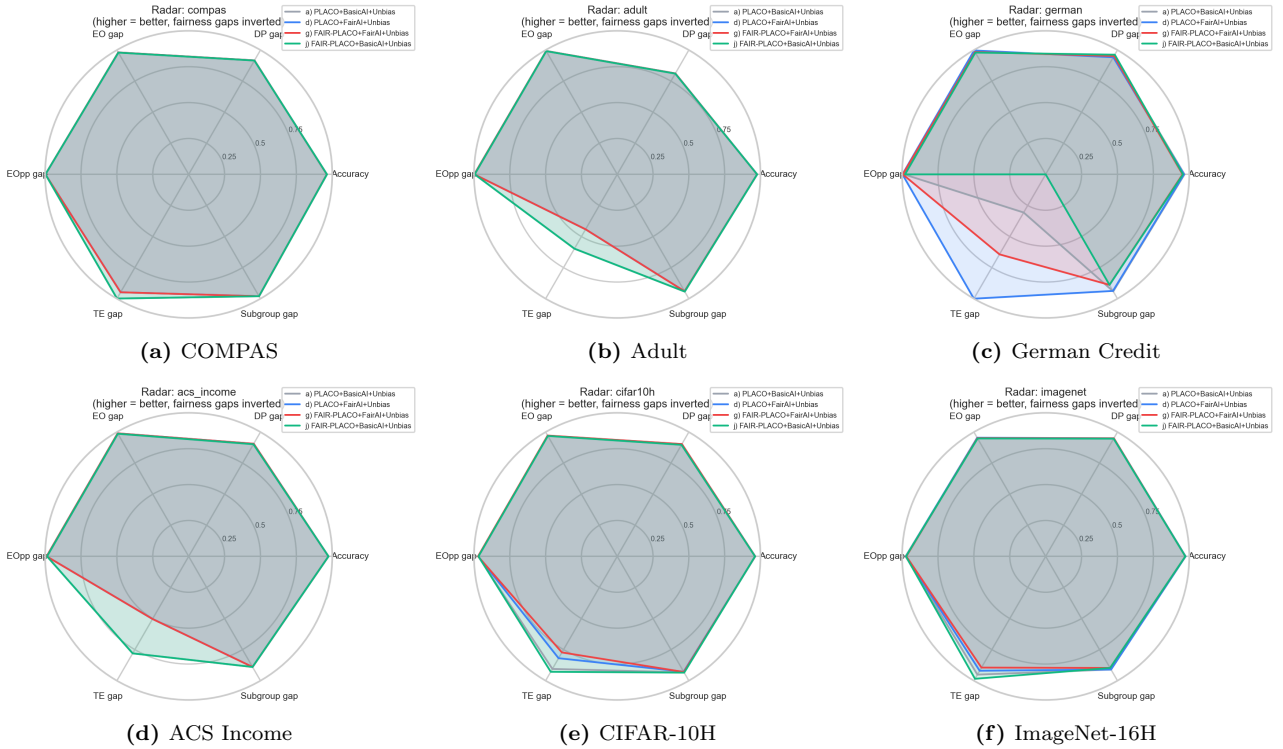


Figure 1: Per-dataset radar charts for settings a, d, g, j. All axes are scaled so higher = better; fairness gap axes are inverted.

10 Conclusion

This report introduced FAIR-PLACO, a framework that brings fairness awareness into the Human-AI team formation problem. Two complementary mechanisms are proposed: a fairness-constrained AI model that reduces the group fairness gap in the probabilistic prior, and a Lagrangian value function that penalises marginal increases in fairness violations during human subset selection.

Experiments across 12 settings, 6 datasets, and 3 human-bias regimes show that:

1. A fair AI prior (FairAI) is the highest-value single intervention, reducing EO by up to 14% with negligible accuracy cost.
2. Lagrangian routing adds complementary fairness gains, especially on Equal Opportunity, and is effective even when used with a standard unconstrained classifier.
3. Human bias is a dominant source of group unfairness that neither mechanism can fully counteract — developing bias-resilient human selection strategies is the most important open direction.
4. Automation-biased humans, when paired with a fair AI, can paradoxically *improve* fairness by propagating the AI’s equalized-odds behaviour, a finding with practical implications for designing human-AI interfaces.

The FAIR-PLACO system provides a modular template: any combination method, fairness metric, and subset selection algorithm can be plugged in, making it straightforward to extend to new domains, richer bias models, and more nuanced fairness definitions as the field evolves.

References

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